Projective Reed–Solomon syndromes beyond redundancy four: deep holes, coherent polar flags, and modular carriers

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Abstract

We classify, up to projective-semilinear equivalence, the one-column MDS extensions arising from projective Reed–Solomon codes at redundancy five over every prime power $q \geq 7$, and give their exact projective counts. Equivalently, we classify the projective deep-hole syndrome directions in the first case left open by the classification through redundancy four. A syndrome determines a Hankel linear system of binary forms and is deep exactly when that system contains no completely split squarefree member. The generic redundancy-five case is controlled by an integral off-diagonal fiber-square curve of arithmetic genus one.

For higher redundancy we introduce coherent polar flags: iterated contractions in which every selected factor remains a marked forbidden root. After separating the level-specific contained-component problem from the transverse point-count argument, this method gives complete all-field classifications at redundancies six and seven; the redundancy-seven split-free list is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. The three-marker argument at redundancy eight leaves only the persistent families for $q \geq 43$; the four-marker and binary-quartic slice arguments give the analogous conclusion at redundancy nine for $q \geq 53$.

In characteristic two we determine the geometric degeneracy strata of an ordered Hessian pullback, its complete root-compatible carrier, and an effective base-selection bound. We also analyze the power-of-two Lucas arithmetic and the distinguished degree-nine e_7 orbit. The remaining degree-nine carrier strata and redundancy-ten deep holes are not classified.

1 Introduction

The principal coding output is an exact classification, up to projective-semilinear equivalence, of the one-column MDS extensions of the projective Reed–Solomon codes treated below, together with exact projective extension counts. We express this through deep-hole syndromes because that language exposes the Hankel geometry driving the proof.

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \dots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \dots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [Kai17, ZWK20]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [ZWK20]. Our first theorem gives the corresponding all-field redundancy-five classification.

Table 1: Notation used throughout the paper.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \mathrm{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
b, c	transverse bad-carrier and marker-collision degrees
$d = 2^s$	power-of-two Lucas degree; used only in Section 9

Definition 1.1 (Working terminology)**.** A *carrier* is a closed subvariety of a syndrome space on which the generic transverse splitting argument degenerates. A *persistent carrier* is a catalecticant component propagated by ordinary contraction; a *modular* or *Lucas carrier* is a component created by a positive-characteristic contraction kernel. A direction is *shallow* when it has a split squarefree witness. A pullback is *root-compatible* when the chosen contraction factor remains a marked forbidden root on every lower incidence. An *ordered-root cover* remembers an ordering of the roots over the squarefree splitting locus. A field is *code-admissible* for a displayed statement when it satisfies both that statement's field bound and its covering-radius hypothesis.

Theorem 1.2 (Complete redundancy-five classification)**.** Let $q \geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;

(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;

(iii) if $\mathrm{char}\,\mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;

(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;

(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table 4, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\mathrm{char}\,\mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3. Cyclic descent gives the arithmetic families in Theorem 1.2; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [AP95] forces a split fiber for $q \geq 23$. The certified finite computations in Section 10 handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a *polar flag*; every chosen root remains marked because a lower witness containing it would lift with multiplicity two. The pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has three outputs.

Theorem 1.3 (Classification results and explicit gates beyond five)**.** The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit.

(b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section 6. Together with the structural theorem, these finite classifications give the all-field result: for every $q \geq 13$ the split-free locus consists of the persistent families and, when $q = 2^m$ with m odd, one central nucleus point. This is a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$.

(c) For $q \geq 43$, the projective deep-hole directions of $\mathrm{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families. The same conclusion holds for $\mathrm{PRS}(q-8)$ when $q \geq 53$. In either case the asserted families have total size $q(q+1)^2/2$.

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius,}$$

where $T = \{z \in \mathbb{F}_{q^2}^{\times} : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char} \mathbb{F}_q$ divides $r-1$.

The three parts have different scopes. Parts (a) and (b) are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$. Part (c) is unconditional in its two displayed high-field ranges. Neither clause makes a classification claim below its threshold.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [Kai17, ZWK20]; the covering-radius input in the high-rate range is due to Seroussi and Roth [SR86]. The broader arc/MDS and code-automorphism background is surveyed or established in [BL19, Dür87]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [ZW17, XHX17, Xu23]; the recent extension framework of [WDC23] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [IK99, CS11]; our use of divided powers keeps the small-characteristic contraction formulas explicit. Binary-quartic orbit theory in characteristic different from two and three [KPP25], factorization statistics in linear families [CMP17], normal-rational-curve nuclei [GH00], and the general Frobenius/monodromy semantics of splitting families [Wan26] are inputs. The first two references identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Standard function-field, Artin–Schreier, and linearized polynomial background is taken from [Sti09, LN96]; the characteristic-two orbit and component exhaustions themselves are proved internally, not imported from the quartic-orbit reference. Accordingly, this paper's contributions are separated into the redundancy-five through seven classifications, the conditional transverse induction mechanism, the fixed-level high-field applications, and the characteristic-two contained-carrier results. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger.

Organization. Section 2 first gives a proof-free guide to the argument, and Section 3 gives the Hankel dictionary. Section 4 proves Theorem 1.2. Sections 5–7 develop coherent polar induction and prove Theorem 1.3. Section 8 gives the characteristic-two ordered-Hessian replacement. Section 9 treats Lucas-carrier arithmetic. Section 10 states the computational and formal trust boundary, and Section 11 records the exact open boundary.

2 Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

Syndrome to splitting system. A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \mathrm{Sym}^{r-2} E^\vee$. A product of $r-2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

The redundancy-five base cover. At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2,2)$ curve. This is the bottom cover used at every later level.

Why independent lower fibers fail. At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

Coherent replacement and the fork. Choose a prospective root λ, contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad carrier and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: the persistent and modular components require a separate level-specific classification.

Lifting. On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, old-marker, new-marker, and self-collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f. Figure 1 records this sequence and the contained/transverse fork.

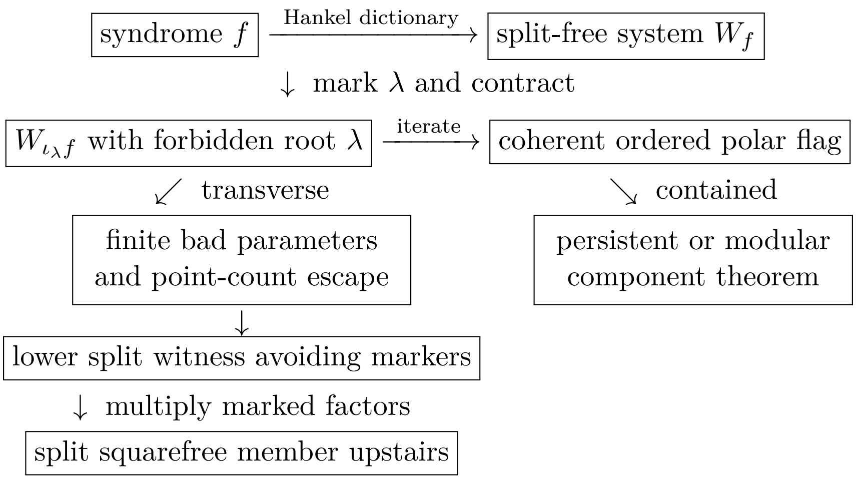

Figure 1: Information flow of coherent polar induction. Solid arrows are the statements proved in the degrees treated here. The contained branch is not a consequence of the transverse sieve.

Table 2: Mathematical dependency map.

Result	Printed geometric argument	Imported input
R5	Hankel dictionary; gcd strata; cyclic descent; integral $(2,2)$ fiber square and point bound	lower tangent/secant families; high-rate radius theorem; singular-curve bound
R6	one-marker polar line; ordinary and binary contained components; transverse escape	R5 lower package and radius gate
R7	two-marker contraction; contained rank-two theorem; transverse escape	R5/R6 lower packages; separate covering-radius result
R8	three-marker cover; explicit contained and collision calculations; point-count budget	proved lower packages
R9	four-marker cover; six-slice identity; characteristic-seven carrier; point-count budget	proved lower packages; quartic normal forms where invoked
Modular	ordered-Hessian strata; root-compatible pullback; linearized and Artin–Schreier covers	normal-rational-curve nuclei and standard finite-field theory

Table 3: Conclusions, electronic evidence, and scope boundaries. The radius gate is logically separate from the split-free classification, and each certificate closes only its displayed finite or algebraic domain.

Result	Conclusion and radius gate	Evidence and boundary
R5	complete deep-hole classification for every $q \geq 7$	The census, canonical representatives, stabilizers, Frobenius links, and independent replay close the finite bridge. They do not supply the all-field cubic-cover argument or radius theorem.
R6	complete deep-hole classification for every $q \geq 7$	The finite census and normal-form reconstruction do not replace the contained/transverse proof or all-field continuation.
R7	complete split-free classification; deep where $\rho = 6$, in particular $q \geq 11$	The split-free census closes the finite bridge, not the structural continuation or covering-radius promotion.
R8	persistent-family classification for $q \geq 43$	The algebra and nucleus regression are not an ambient census or a replacement for the printed component theorem.
R9	persistent-family classification for $q \geq 53$	Residual and R9-49 replays check bounded algebra; Lean checks the synthesis implication and integer budgets. Integrality, rational points, PRS identification, exhaustion, and the group action remain printed mathematics.
Modular	component and e_7 results only; no R10 classification	The Hessian, Lucas, and e_7 regressions do not replace the geometric component proofs.

Referee roadmap. The shortest route to the headline R5 theorem is Lemma 3.1 and Corollary 3.2, followed by the radius gate and gcd strata in Propositions 4.1–4.3, the cubic incidence in Proposition 4.4, the cyclic and S_3 analyses in Lemmas 4.5–4.6, and the finite bridge in Proposition 4.7. Tables 4 and 5 make the bounded-field output locally identifiable and numerically auditable.

The later R6 and R7 proofs use the R5 splitting cover as their bottom package; R8 and R9 then use those proved lower packages recursively. Section 5 may be read as the abstract mechanism linking these levels. The ordered-Hessian and Lucas analyses in Sections 8 and 9 are logically separate after the dictionary and can be read independently of the R8/R9 point counts. Running the supplement is required only to reproduce the finite bridges, canonical orbit records, and regression or formal checks identified in Table 3; it is not a substitute for any printed geometric argument.

The proofs now follow these arrows in order: dictionary, redundancy-five base cover, polar construction, contained components, transverse point count, and fixed-level applications. This is a scope boundary, not a second theorem inventory.

3 The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. The perfect divided-power/symmetric-power pairing is characterized by coefficient extraction. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

Lemma 3.1 (Hankel criterion)**.** Let $\lambda_1, \dots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \dots, \nu(\lambda_{r-2})\big) \quad \Longleftrightarrow \quad g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, g is the characteristic polynomial of an order-$(r-2)$ recurrence. The length-r solutions are spanned by the geometric sequences $\nu(\lambda_i)$; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary 3.2 (Split-free criterion)**.** Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We call the latter property *split-free* even when the covering-radius hypothesis has not been established. Thus every deep syndrome in the radius-$r-1$ range is split-free, while a small-field split-free census is not silently promoted to a code deep-hole classification.

The persistent locus is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r-3$,

$$r-3 \le r-1-\deg\gcd(W_f),$$

so its degree is at most two.

3.1 Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is an uncovered point of the normal rational curve and hence gives a one-column MDS extension whenever the stated covering-radius gate applies. The classification theorems therefore give exact projective deep-hole counts, not only normal forms. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects a modular nucleus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table 4 and recorded in the supplement.

For each fixed redundancy treated here, constructing and row-reducing the Hankel matrix and performing the listed family-membership tests costs $O(r^3)$ field operations. This count treats factorization of the displayed bounded-degree forms and access to the exceptional orbit tables as fixed-degree work. It excludes field construction, bit complexity, and the one-time generation and certification of those tables; an implementation may also charge separately for its chosen orbit-normalization routine. Thus the bound describes the online linear-algebra stage at the fixed redundancies proved here. It is not an end-to-end uniform arbitrary-r complexity claim, a general decoder, or a preprocessing bound. A direct witness search would instead inspect the projective members of the $(r-3)$-dimensional system W_f and factor degree-$(r-2)$ forms; coherent polar induction is the mechanism that replaces that ambient search in the levels proved below.

4 Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition 4.1 (Radius gate)**.** For every prime power $q \geq 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. Seroussi–Roth prove completeness when

$$k \geq \left\lfloor \frac{q-1}{2} \right\rfloor$$

apart from the even-field dimensions $k = 2, q-2$ [SR86]. Here $k = q-4$, the inequality holds for $q \geq 7$, and neither exceptional dimension occurs. Thus Corollary 3.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

4.1 Gcd strata

Proposition 4.2 (Quadratic gcd)**.** If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U\rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma 3.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [Kai17, ZWK20]. □

Proposition 4.3 (Linear gcd)**.** Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \geq 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \to \mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t, \iota(t)) : t \in \mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0, \iota(\lambda_0))$ and $(\iota(\lambda_0), \lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q \geq 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2\rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. □

4.2 The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition 4.4 (Cubic incidence and off-diagonal fiber square)**.** The curve X_f is geometrically integral of bidegree $(1, 3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2, 2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0, e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1, 1)$ from the fiber square leaves bidegree $(2, 2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. □

Lemma 4.5 (Cyclic stratum)**.** Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0 : 0 : 1 : 0 : 0)$, whose pencil is $\langle T^3, U^3 \rangle$; this fixed point is deep.

(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2 \rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2 - 1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3 \rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0,0,1,3t,6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3 \rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2 U + \beta T U^2 \rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - a e_4, \qquad W_{f_a} = \langle U^3, T^3 + a T U^2 \rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2-1)/2$. □

Lemma 4.6 (S_3 stratum)**.** If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition 4.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. Aubry–Perret gives

$$\#Y_f(\mathbb{F}_q) \geq q - 2\sqrt{q}.$$

The diagonal costs

$$Y_f \cdot \Delta = (2,2) \cdot (1,1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q - 2\sqrt{q} > 12$, a rational off-diagonal unramified point remains. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. The inequality holds for every prime power $q \geq 23$. □

Proposition 4.7 (Exhaustion and finite bridge)**.** The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3-stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions 4.2 and 4.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas 4.5 and 4.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. □

Table 4: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \dots, a_4]$ is in the divided-power syndrome coordinates of Section 3. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	`[0,1,0,0,3]`	28	12	–
7	`[0,1,0,0,2]`	112	3	–
7	`[0,0,1,0,3]`	168	2	–
7	`[0,1,0,3,2]`	168	2	–
7	`[1,0,0,2,1]`	168	2	–
8	`[0,1,1,0,2]`	252	2	$\to [0,1,1,0,4]$
8	`[0,1,1,0,4]`	252	2	$\to [0,1,1,0,6]$
8	`[0,1,1,0,6]`	252	2	$\to [0,1,1,0,2]$
9	`[1,0,0,1,2]`	180	4	–
9	`[0,1,0,4,1]`	360	2	$\leftrightarrow [0,1,0,4,4]$
9	`[0,1,0,4,4]`	360	2	$\leftrightarrow [0,1,0,4,1]$
11	`[1,0,0,1,10]`	330	4	–
11	`[1,0,0,1,1]`	660	2	–
13	`[0,1,0,0,4]`	182	12	–
13	`[1,0,0,4,7]`	546	4	–
17	`[1,0,0,1,5]`	1224	4	–
19	`[0,1,0,0,4]`	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table 4 are therefore visibly distinguished without requiring a working-directory file.

Table 5: Independent R5 completeness totals. The first equality is the direct syndrome enumeration versus the orbit–stabilizer sum; the second decomposes the same total into the formula in Theorem 1.2 and the sporadic rows above.

q	direct = orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

5 Coherent polar induction

5.1 Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g\rangle = \langle f, \lambda g\rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad\Longleftrightarrow\quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition 5.1 (Contraction and marker spaces)**.** Let $C_f : E^\vee \to \Gamma^{n-1}E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \dots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \dots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \dots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]\big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - r a_0, \dots, a_n - r a_{n-1}), \qquad \Phi_f(\infty) = (a_0, \dots, a_{n-1}).$$

Example 5.2 (A complete contraction at $q = 29$)**.** Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle .$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1), \qquad h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (1) therefore lifts h to

$$t\, h(t) = (0, 4, 9, 28, 5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface used later: Hankel kernel, persistent test, marked contraction, lower factorization, marker avoidance, and squarefree lifting.

Definition 5.3 (Lower splitting package)**.** At level (n, j) a lower splitting package consists of:

(i) a finite stratification $X_{n-j}^{\mathrm{split}} = \coprod_\alpha X_\alpha$ of the ordered squarefree splitting incidence over S_{n-j};

(ii) for each α, a specified identity or Frobenius-twisted geometrically integral root cover $Y_\alpha \to X_\alpha$, with genus bound g_α and point bound

$$\#Y_\alpha(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

(iii) a closed bad scheme $B_{n-j} \subset S_{n-j}$ containing the loci on which the chosen cover is not available;

(iv) branch, diagonal, old-marker, new-marker, and self-collision divisors whose total degree on Y_α is at most δ_α;

(v) bounds b and c for the pullbacks of B_{n-j} and the finite self-collision divisor to a non-contained first-polar line.

The index α names a geometric stratum together with its constant field and Frobenius twist. All degrees are geometric degrees after the displayed pullback. “Computed” means that equations for B_{n-j} and every divisor are obtained by finite contraction, elimination, factorization, and rank tests over the coefficient ring.

The construction is summarized in Figure 2. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

$$\begin{array}{cccc} [f] \in S_n & \dashrightarrow \quad \Phi_f(U_f) \subset S_{n-1} & \longrightarrow & \text{lower ordered splitting cover} \\ & \searrow \text{ contained} & & \text{persistent/Lucas component,} \\ & \nearrow \text{ transverse} & & \text{delete branch, diagonal, and markers} \\ & & & \downarrow \text{ lift by (1)} \\ & & & \text{split squarefree witness in } W_f. \end{array}$$

Figure 2: Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem 5.4 (Polar-flag construction)**.** The maps of Definition 5.1 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j}\cdots\iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. □

Definition 5.5 (Contained-component assertion)**.** For a specified lower package, $\mathrm{CC}(n, j)$ is the scheme-theoretic assertion that every $f \in S_n$ for which $\Phi_f^* B_{n-j}$ or a marker-collision section vanishes identically belongs to the explicitly listed closed subscheme $\mathcal{P}_n \cup \mathcal{M}_n$. An identically zero collision differential is part of $\mathcal{M}_n$; it is never counted as a finite divisor.

This is a level-specific theorem obligation, not an assumption hidden inside the point-count argument. Sections 6 and 7 state the contained-component calculations used at redundancies six through nine. No assertion $\mathrm{CC}(n, j)$ is made for an arbitrary degree.

Theorem 5.6 (Effective transverse induction)**.** Fix a lower splitting package as in Definition 5.3 and suppose $\mathrm{CC}(n,j)$ has been proved for that package. Put

$$\mathcal{H}_\kappa(g,\delta) = \left\lfloor \left(g + \sqrt{g^2+\delta-\kappa} \right)^2 \right\rfloor + 1$$

and assume $\delta \geq \kappa - g^2$. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha,\delta_\alpha) \quad \text{and} \quad q+1 > b+c,$$

then every split-free Hankel system belongs to the contained scheme $\mathcal{P}_n \cup \mathcal{M}_n$.

Proof. For a system outside $\mathcal{P}_n \cup \mathcal{M}_n$, the assertion $\mathrm{CC}(n,j)$ makes the pulled-back bad and self-collision schemes finite of total degree at most $b+c$. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q+1$, choose a rational contraction parameter outside them. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_\alpha - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding every marker. Equation (1) lifts it to a split squarefree member upstairs, contradicting split-freeness. □

Remark 5.7 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. The singular cubic fiber square of Lemma 4.6 uses $\kappa = 0$. The normalized genus-one covers in the fixed-level applications use $\kappa = 1$. Thus the redundancy-eight triple $(g,\delta,\kappa) = (1,30,1)$ gives the real cutoff $(1+\sqrt{30})^2 < 42$, whose first prime power is 43; the redundancy-nine triple $(1,36,1)$ gives the strict cutoff $q > 49$, whose first prime power is 53.

Table 6: Normalized point-count thresholds used in this paper.

Application	g	δ	κ	integer lower bound	first prime power
R5 cubic fiber square	1	12	0	$q \geq 22$	23
R6 marked cover	1	18	1	$q \geq 28$	29
R7 marked cover	1	24	1	$q \geq 35$	37
R8 three-marker cover	1	30	1	$q \geq 42$	43
R9 four-marker cover	1	36	1	$q \geq 50$	53

5.2 Contained-component calculations

The ordinary rank-two carrier admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition 5.8 (Contained rank-two polar lines)**.** Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \mathrm{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1}E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f

to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \le 2\} \quad \Longleftrightarrow \quad \operatorname{rank} A \le 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C^{(2)}_{\iota_\lambda f} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n - 4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \ge (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E^\vee_{\bar{k}})} \lambda \operatorname{Sym}^{n-3} E^\vee_{\bar{k}} = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C^{(2)}_{\iota_\lambda f}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the non-contained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. $\square$

Thus Proposition 5.8 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant carriers. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular carriers are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E/\widetilde{\mathcal{N}}) \right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census.

Remark 5.9 (Scope of the contained calculations)*.* The modular contraction kernel is an exact finite linear-algebra construction. Proposition 5.8 supplies the ordinary persistent component uniformly, but other lower bad schemes and marker collisions still require level-specific calculations. Neither the proposition nor Theorem 5.6 classifies those additional components.

6 Redundancies six and seven: complete and gated results

6.1 Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition 6.1 (Persistent net and orbit law)**.** If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The deep persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. $\square$

Proposition 6.2 (First-polar line and secant degree)**.** For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012} r^3 + \Delta_{013} r^2 - \Delta_{023} r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \le 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. $\square$

Proposition 6.3 (Cyclic and collision degrees)**.** For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;

(ii) the pointed ramification condition has degree at most six and does not vanish identically;

(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most eighteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{\operatorname{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee)\right\}.$$

When 2 and 3 are invertible this is the Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition 6.4. For a basis F_0, F_1, F_2 of the $g^2_4 = W_f$, pointed collision is the rank-one condition

$$\operatorname{rank}\begin{pmatrix} F_0(r) & F_1(r) & F_2(r) \\ F_0'(r) & F_1'(r) & F_2'(r) \end{pmatrix} \le 1.$$

Its Wronskian divisor has degree six. In characteristic at least five it cannot vanish identically because the order sequence is classical. In characteristic three an identically zero first-derivative minor would make the net a pullback from a series of degree at most one, hence of vector dimension at most two. In characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4\rangle$. If two consecutive Hankel rows had this common kernel, comparison of the overlapping coefficients would force $a_0 = \cdots = a_5 = 0$. Thus some derivative minor is nonzero in both modular characteristics. If instead the derivative row is everywhere proportional to the value row, the corresponding rational map is inseparable; after cancelling its pencil gcd, the same dimension argument places it on the fixed-factor boundary. Hence the ramification scheme is a genuine divisor of degree at most six.

Finally the R5 off-diagonal curve has the original deletion degree twelve. The unique cubic through a specified marker contributes at most six ordered pairs, giving $\delta = 18$ and

$$q - 2\sqrt{q} > 18.$$

Its first prime-power solution is 29. □

Proposition 6.4 (Modular contained components)**.** In characteristic three the lower wild carrier is the degree-four cone

$$\operatorname{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. $\square$

Proposition 6.5 (Binary nucleus arithmetic)**.** The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q+1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

The root differences of $A+Bt+Ct^4$ lie in the kernel of $u \mapsto u^4+(B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q-1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. $\square$

The covering-radius gate is complete: Seroussi–Roth gives $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 9$, while Certificate R6 checks it directly at $q = 7, 8$.

Theorem 6.6 (Redundancy six)**.** For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent stratum of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;

(ii) one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \geq 5$.

There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five.

Proof. Propositions 6.1–6.5 classify the common-factor and contained components and exclude a trivial-gcd exception in the conceptual ranges: characteristic at least five from $q \geq 29$, characteristic three from its next unchecked field $q = 81$, and characteristic two outside $\mathcal{N}_3$ from $q \geq 32$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. $\square$

The five small tables are exhaustive computations. Their semilinear orbit counts are $18, 5, 2, 2, 1$; a collision-energy invariant plus, in odd characteristic, the quintic root-divisor type separates the classes up to precisely Frobenius fusion. Certificate R6 supplies a canonical representative, stabilizer, member histogram, and Frobenius link for every projective orbit, while Certificate R6-NF supplies the collision-energy and root-divisor fingerprint. Orbit counts alone are not used as identifiers.

6.2 Redundancy seven

For

$$f = (a_0 : \dots : a_6) \in \mathbb{P}(\Gamma^6 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\ a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r) = (a_1 - ra_0, \dots, a_6 - ra_5).$$

Coefficient collection gives

$$(t-r)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

Proposition 6.7 (Two-marker lower cover)**.** Let x be a forbidden root on a trivial-gcd quartic net. On the generic S_3 bottom fiber, the diagonal and branch divisors and the two marked-root fibers have total degree at most

$$12 + 6 + 6 = 24.$$

Hence $q - 2\sqrt{q} > 24$, and in particular every prime power $q \geq 37$, produces a split lower quartic avoiding x.

Proof. The R5 off-diagonal curve is geometrically integral of arithmetic genus one and has base deletion degree twelve. Each marker belongs to a unique cubic fiber, which contains at most six ordered pairs. Delete those two fibers and apply the $\kappa = 0$ bound. ☐

Proposition 6.8 (Exact-gcd-one avoidance)**.** Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three and $x \neq r$. For $q \geq 16$ it contains a split cubic avoiding both x and r.

Proof. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u,v). None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\operatorname{Sym}^3 E^\vee$. A nonzero bidegree-$(a,b)$ divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. ☐

Proposition 6.9 (Marked self-collision)**.** After fixed factors are removed, the self-collision divisor of the moving g^3_5 has degree at most eight.

Proof. It is the ramification divisor of a separable g^3_5, whose degree is $(3+1)(5-3) = 8$. If the series were inseparable, it would factor through Frobenius and lie in the p-th-power subspace of quintics; that subspace has dimension at most three, smaller than the four-dimensional Hankel series. ☐

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition 6.10 (Central binary lift)**.** The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3\rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5\rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition 6.5. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. □

Corollary 6.11 (Degree-six contained component)**.** The assertion CC(6, 1) for the lower rank-two carrier holds in every characteristic. Its contained sextics have upper catalecticant rank at most two; after the shallow rank-one and rational secant loci are removed, they are exactly the persistent tangent and conjugate-secant families.

Proof. Apply Proposition 5.8 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition 6.10. □

Theorem 6.12 (Redundancy-seven classification)**.** Certificate R7 unconditionally classifies every prime power $q \le 32$ in its displayed domain. For every prime power $q \ge 13$, the split-free syndromes at redundancy seven are the persistent stratum, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles are:

$$\begin{array}{r|l} 7 & 56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141} \\ 8 & 63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86} \\ 9 & 180^3,\ 240^6,\ 360^{18},\ 720^{27} \\ 11 & 264^2,\ 440,\ 660^2. \end{array}$$

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\mathrm{PRS}(q - 6)) = 6$ for every prime power $q \ge 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For $q \ge 37$, Propositions 6.7–6.9 give the transverse witness after at most three catalecticant intersections, eight self-collisions, and, in characteristic two, one intersection with the nucleus line. Corollary 6.11 leaves exactly the persistent locus and Proposition 6.10. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \ge 11$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records.

The $q = 19$ lower net $\langle 1, t^3, t^4\rangle$ has six split members, all containing the marked point at infinity, yet no coherent sextic polar line lies in its orbit. This is the concrete reason Theorem 5.6 is formulated for parameterized flags rather than a set of bad lower fibers.

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \ge 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

Table 7: Field-range closure ledger. "Certificate" means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 16, 17, 19$	direct census and independent replay
	$q \geq 23$	cubic-cover geometry, Lemma 4.6
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

7 Fixed-level high-field theorems

7.1 Redundancy eight

Proposition 7.1 (Generic three-marker bottom cover)**.** After three contractions, the generic bottom object is the R5 off-diagonal $(2,2)$ curve. The diagonal and branch deletion costs twelve and three marked roots cost eighteen, so $\delta = 30$. The inequality

$$q + 1 - 2\sqrt{q} > 30$$

holds at every prime power $q \geq 43$. The first-polar line has transverse/collision budget

$$3 + 1 + 10 = 14,$$

where ten is the ramification degree of a moving g^4_6.

Proof. Markers do not change the bottom curve or its geometric monodromy; each removes at most the six ordered pairs in its unique cubic fiber. The point bound is the $\kappa = 1$ row of Table 6. A noncontained line meets the lower rank-two catalecticant carrier in degree at most three, and the lower binary central point in at most one parameter. After fixed factors are removed, self-collision is ramification of a separable g^4_6, of degree $(4+1)(6-4) = 10$. Separability follows because the p-th-power subspace has dimension smaller than the five-dimensional series. □

Definition 7.2 (Three-marker lower-package gate)**.** The assertion $\mathrm{LP}(6,1)$ says that the pointed R7 splitting incidence, after the persistent rank-two carrier, the binary central point, and the marked-collision divisor are removed, has the geometrically integral identity twists and deletion bounds used in Proposition 7.1. It includes equations and monodromy checks for the recursive cyclic, wild, inseparable, gcd, and branch strata; a finite-field regression is not a substitute for those geometric data.

Lemma 7.3 (Bottom strata)**.** For the two-step contraction of a pointed sextic, the lower rank, gcd-one, cyclic, wild, inseparable, and branch strata are exactly the closed schemes displayed below. The reversed homogeneous chart introduces no additional stratum.

Proof. For a pointed sextic $f = (a_0, \ldots, a_6)$ with old marker x, contract successively:

$$b_i(r) = a_{i+1} - ra_i, \qquad c_i(r,s) = b_{i+1}(r) - sb_i(r).$$

For $c = (c_0, \ldots, c_4)$ put

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}.$$

On the rank-two-kernel chart, if p_0, p_1 span $\ker H_c$, the bottom ordered-root curve is

$$G_c(u,v) = \frac{p_0(u)p_1(v) - p_1(u)p_0(v)}{[u,v]} = 0. \tag{2}$$

The rank-at-most-two secant carrier, including the gcd-two and rank-drop boundary strata, is

$$D(c) = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix} = 0. \tag{3}$$

For an exact common root z, the gcd-one incidence is cut out by

$$\operatorname{rank} \begin{pmatrix} H_c & & & \\ 1 & z & z^2 & z^3 \end{pmatrix} \leq 2. \tag{4}$$

with the evident homogeneous row at infinity. After substitution, (3) has degree at most three in s, while each 3-minor in (4) has degree at most two for fixed z.

The remaining reducible-monodromy carriers are explicit. Put $z_i = \binom{4}{i} c_i$. Outside characteristics two and three the cyclic closure is

$$[u : v : w] \longmapsto [u^2 : 2uv : v^2 + 2uw : 2vw : w^2], \tag{5}$$

a degree-four projected Veronese surface containing no line. In characteristic three it is the degree-four cone $\bowtie (e_2, C_4)$; its only lines are rulings, and the consecutive-row overlap excludes every ruling as a polar line. In characteristic two it is

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3 \rangle, \qquad c_0 = c_4 = 0. \tag{6}$$

whose unique contained polar-line component is $\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3 \rangle$. The inseparable degeneration is the derivative-minor locus of p_0, p_1 and introduces no further stratum. Outside characteristics two and three a nonconstant degree-three map cannot factor through Frobenius. In characteristic three an inseparable cubic pencil is projectively $\langle X^3, Y^3 \rangle$; the two Hankel rows then force $c_0 = c_1 = c_3 = c_4 = 0$, the vertex e_2 of the wild cone. In characteristic two, after cancelling a nonzero pencil gcd, an inseparable ratio has degree at most two and is a rational function of t^2. Homogenizing its numerator and denominator to cubics supplies the same linear factor to both, so it belongs to the exact-gcd-one boundary. Finally, the branch equation on this chart is $p_0' p_1 - p_0 p_1' = 0$. These carrier and branch loci are the scheme-theoretic images of the displayed incidences (equivalently their Fitting-minor ideals); the reversed homogeneous chart supplies infinity. Thus these are closed strata over the coefficient ring, not a classification only of their rational points. □

Lemma 7.4 (Bottom monodromy and deletion)**.** Off the strata of Lemma 7.3, the bottom ordered-root curve is geometrically integral of arithmetic genus one and has deletion degree at most 30. On the exact-gcd-one stratum, the residual deck graph supplies a squarefree pair avoiding the three markers for $q \geq 43$.

Proof. Off (3)–(6), the separable cubic map is noncyclic, so its geometric monodromy is S_3. Transitivity on ordered pairs of distinct roots makes (2) geometrically integral. It has arithmetic genus one. The diagonal costs four, the degree-four different costs at most eight off-diagonal ordered pairs, and each of the three marker fibers x, r, s costs at most six. Thus

$$\delta \leq 4 + 8 + 3 \cdot 6 = 30. \tag{7}$$

Hasse–Weil supplies a surviving rational point for every $q \geq 43$.

It remains to show that the strata removed above do not hide another component. On an exact-gcd-one bottom stratum write the cubic pencil as $\ell_z Q$, with Q a basepoint-free pencil of quadratics. In the separable case its deck involution ι is rational, so its graph has $q+1$ rational points. At most two are fixed and at most two ordered graph points involve each of z, x, r, s. A point remains for $q \geq 43$, giving three distinct roots outside the markers. The inseparable degree-two case is on the rank/boundary carrier. ☐

Lemma 7.5 (Two-marker selection)**.** Fix two old markers x, r. Outside the declared contained strata, the bad divisor on the internal s-line has degree at most 19. For $q \geq 43$ one can choose s so that the lower package supplies a split squarefree quartic avoiding x and r.

Proof. The complete divisor table on the s-line is

failure	defining mechanism	degree
rank at most two	$D(c(s))$	3
cyclic/wild monodromy	pullback of (5)–(6)	4
new gcd root s	$\mathrm{Ram}(g^2_d),\ d \leq 4$	6
gcd root x or r	one nonzero minor in (4)	$2+2$
marker collision	$s = x,\ s = r$	2

In characteristic two the cyclic entry has degree one, so degree four is a uniform bound. The total is

$$3 + 4 + 6 + 2 + 2 + 2 = 19 < q + 1. \tag{8}$$

The rank pullback cannot be the whole line: Proposition 6.2 identifies that containment exactly with quadratic gcd of the original net. The only possible contained cyclic line is the declared binary nucleus. For a fixed marker, not all minors in (4) can vanish identically without giving the original quartic net that fixed factor: otherwise the hyperplanes $W \cap \ker(\mathrm{ev}_s)$ and $W \cap \ker(\mathrm{ev}_x)$ agree for dense s, making the evaluation map constant. Choose one nonzero quadratic minor. For self-collision, take a generic pencil projection of the moving g^2_d, $d \leq 4$. It is separable: in characteristic two the only dimension-three equality case is the pure-square space $\langle 1, t^2, t^4\rangle$, which the consecutive Hankel overlap equations do not realize; all other p-power spaces have smaller dimension. Riemann–Hurwitz gives degree at most $2d-2 \leq 6$, and every self-collision maps into this ramification divisor. Choosing s outside this divisor and applying the preceding gcd-zero or gcd-one bottom analysis gives a split squarefree quartic avoiding the two old markers. This proves the two-old-marker lemma. ☐

Lemma 7.6 (Outer marker selection)**.** For a pointed sextic outside the persistent and modular contained strata, the bad divisor on the outer first-marker line has degree at most 15. A surviving outer marker lifts the witness of Lemma 7.5 to a split squarefree quintic avoiding the original marker.

Proof. The lower rank-two pullback costs three parameters and the binary nucleus costs one. If a contracted net has exact gcd ℓ_z, the condition $z = r$ maps into the ramification of a generic separable pencil projection of the moving g^3_d, $d \leq 5$. Riemann–Hurwitz bounds it by $2d-2 \leq 8$. The condition $z = x$ is again (4), now with the quartic evaluation row, and costs at most two because not all of its quadratic minors vanish off the fixed-marker boundary, by the same evaluation-hyperplane argument. Together with $r = x$, the internal divisor has degree

$$3 + 1 + 8 + 2 + 1 = 15. \tag{9}$$

If the contracted net has trivial gcd, the two-old-marker lemma applies. If it has exact gcd with $z \notin \{x, r\}$, the ordered-pair argument has three $(1,1)$ and two $(2,1)/(1,2)$ bad equations, at most $12(q+1) < (q-2)(q-3)$. The cases $z = x, r$ are precisely the excluded quadratic and ramification divisors. Lifting first by ℓ_s and then by ℓ_r gives a split squarefree quintic avoiding x. $\square$

Proposition 7.7 (Pointed lower package)**.** The assertion $\mathrm{LP}(6,1)$ holds. On the outer first-marker line its exceptional divisor has degree at most fifteen; on the recursive second-marker line the corresponding divisor has degree at most nineteen. Its only positive-genus identity twist is the geometrically integral off-diagonal $(2,2)$ curve of genus at most one with deletion degree 30.

Proof. Lemmas 7.3 and 7.4 identify the bottom strata and the unique positive-genus identity twist. Lemmas 7.5 and 7.6 give the two parameter-line budgets and the coherent squarefree lift. For auditability, the complete exhaustion is summarized here.

Stratum	Defining equation	Degree	Contained case	Transverse conclusion
rank at most two	(3)	3 in the marker	quadratic-gcd persistent carrier	finite intersection of degree at most 3
cyclic or wild	(5) or (6)	at most 4	only the binary nucleus line	finite intersection of degree at most 4
exact gcd one	minors in (4)	2 per fixed marker	fixed-factor boundary	the residual deck graph supplies a squarefree pair
inseparable	derivative minors of p_0, p_1	absorbed above	wild vertex, rank drop, or binary nucleus	no additional stratum
geometric S_3	(2)	bidegree $(2,2)$	only the preceding carriers	integral genus-one cover with deletion degree 30
marker collision	diagonals and ramification	1 per diagonal; at most 6 for self-collision	declared collision boundary	finite divisor on the marker line

The gcd has degree zero, one, or at least two, and a separable gcd-zero cubic cover has monodromy S_3 or C_3; hence the displayed list is exhaustive. No certificate or finite-field regression is used for this integrality assertion. Notice also that the outer and recursive parameter budgets 15 and 19 are already strictly below $q+1$ for $q \geq 19$; the threshold 43 is imposed solely by the genus-one deletion inequality (7), not by a hidden parameter-choice bound. $\square$

Proposition 7.8 (Modular lifts at degree seven)**.** The consecutive-row Lucas calculation leaves no nonzero lift in characteristic two, the line $\mathbb{P}\langle e_2, e_5\rangle$ in characteristic three, and the line $\mathbb{P}\langle e_3, e_4\rangle$ in characteristic five. The latter two lines are shallow in the range $q \geq 43$.

Proof. Lucas' criterion applied to the consecutive binomial rows gives the three stated supports; the exact row reductions are replayed by Certificate R8. In characteristic five, the homogenization of $t^5 - t$ supplies six distinct projective roots and satisfies the two required middle-coefficient equations throughout the lift.

In characteristic three, $\mathrm{PGL}_2(q)$ is transitive on the rational points of $\mathbb{P}\langle e_2, e_5\rangle$, so it suffices to treat e_2. Choose nonzero a, b, c with no two equal up to sign and $a^2+b^2+c^2 = 0$. The nonsingular conic has $q+1$ points; deleting the three coordinate and six equality/antiequality sections costs at most eighteen, so a choice exists for $q \geq 27$. The reciprocal pairs $\pm a^{-1}, \pm b^{-1}, \pm c^{-1}$ give six distinct roots and the two required elementary-symmetric coefficients vanish. $\square$

Proposition 7.9 (Degree-seven contained components)**.** The assertion $\mathrm{CC}(7,1)$ for the structural components of Definition 7.2 holds in every characteristic. The rank-two component lifts to the upper rank-two carrier; the lower binary central point has no nonzero coherent lift; and the marked-collision scheme is never the whole polar line. The only additional nucleus lifts are

$$\mathbb{P}\langle e_2, e_5\rangle \quad \text{in characteristic three,} \qquad \mathbb{P}\langle e_3, e_4\rangle \quad \text{in characteristic five,}$$

and both are shallow in the range $q \geq 43$.

Proof. The rank-two assertion is Proposition 5.8 with $n = 7$. For the binary central point e_3, requiring both consecutive septic rows to be supported at its single coordinate permits a_3 in the first row and a_4 in the second; their overlap is zero. Thus its projective coherent lift is empty. The full nucleus-support overlap is exactly Proposition 7.8, which also supplies the shallow witnesses in characteristics three and five.

It remains to exclude an identically colliding series. Remove the fixed divisor from the five-dimensional W_f and let the moving degree be $d \leq 6$. If its projective map were inseparable in characteristic p, it would factor through Frobenius, so its section space would have dimension at most

$$\lfloor d/p \rfloor + 1 \leq 4,$$

contrary to $\dim W_f = 5$. A generic pencil projection is therefore a separable map of degree at most six. Every marked self-collision is a ramification point of that projection, so Riemann–Hurwitz bounds the collision divisor by $2d - 2 \leq 10$. It is finite and cannot be an additional contained component. ☐

Remark 7.10 (Exact remaining R8 gate). Proposition 7.9 closes the contained-component problem, while Proposition 7.7 closes the pointed lower package. The certificate checks the nucleus overlaps, witnesses, and numerical budgets; the geometric integrality and recursive carrier equations are proved separately above.

Theorem 7.11 (Redundancy eight)**.** For every prime power $q \geq 43$, the deep syndromes of $\mathrm{PRS}(q-7)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$. Their orbit law is T/T^7 modulo inversion and Frobenius. The tangent family splits exactly in characteristic seven.

Proof. Outside the contained locus, Proposition 7.1 chooses a good first contraction, and Proposition 7.7 supplies a three-marker witness. Propositions 7.8 and 7.9 eliminate every nonpersistent contained lift. The imported covering-radius theorem promotes the resulting split-free classification to deep holes. ☐

This theorem makes no assertion about additional deep orbits for $q < 43$.

7.2 Redundancy nine

Definition 7.12 (R9 pointed domain)**.** The assertion $\mathrm{LP}(7, 1)$ is made on the pointed splitting incidence after removing the persistent rank-two carrier, the complete characteristic-specific Lucas-nucleus support, and the fixed-marker and self-collision components. At every recursive level, a polar line contained in one of these supports belongs to that declared closed boundary; the parameter-divisor argument applies only to the noncontained line. Thus a section that vanishes identically is never counted as a finite divisor of its nominal degree.

Proposition 7.13 (Four-marker pointed lower package)**.** The assertion $\mathrm{LP}(7, 1)$ holds for every $q \geq 53$. Its bottom identity twist is the geometrically integral off-diagonal $(2, 2)$ curve of Proposition 7.7, now with four forbidden markers and deletion degree

$$4 + 8 + 4 \cdot 6 = 36. \tag{10}$$

The three recursive parameter-line budgets, from the bottom upward, are at most 22, 18, and 18. At the top degree-eight first-polar line the transverse/component budget is at most

$$3 + 2 + 12 = 17. \tag{11}$$

No geometric-integrality, gcd, or contained-component assertion is included merely as a number in these budgets.

Proof. We retain the notation and the scheme-theoretic strata (2)–(6). Their homogeneous incidence ideals commute with further contraction, so adding a marker changes only the corresponding evaluation and diagonal pullbacks; it does not change the bottom cover. We prove the package in three steps.

Three-pointed quartic net. Let W be a trivial-gcd Hankel net of quartics, let x_1, x_2, x_3 be distinct prescribed markers, and exclude the characteristic-two nucleus already listed in (6). Contract at a fourth parameter s. On the resulting cubic pencil the rank pullback (3), the cyclic/wild pullback (5)–(6), and self-collision have degrees at most 3, 4, and 6, respectively. For each x_i, choose a nonzero quadratic minor in (4); the three fixed-root pullbacks cost at most 6, and the three equations $s = x_i$ cost 3. Thus the parameter divisor has degree at most

$$3 + 4 + 6 + 3 \cdot 2 + 3 = 22. \tag{12}$$

None of these equations vanishes identically. For the rank and cyclic carriers this is the no-contained-line calculation following (8). For a fixed marker, identical vanishing of all the minors in (4) would make the kernels of $\mathrm{ev}_s|_W$ and $\mathrm{ev}_{x_i}|_W$ agree for dense s. The two nonzero functionals would then be proportional, making the projective evaluation map of the three-dimensional trivial-gcd net W constant, a contradiction. For self-collision, a generic pencil projection of the moving g^2_d, $d \leq 4$, is separable by the pure-power dimension argument following (8), and Riemann–Hurwitz gives degree at most 6.

Away from this divisor, a gcd-zero cubic pencil is on the geometric S_3 stratum. Its ordered-pair cover (2) is geometrically integral of genus at most one. The diagonal and different delete at most $4+8$ points, and each of x_1, x_2, x_3, s deletes at most the six ordered pairs in its cubic fiber. This proves (10). If instead the cubic pencil has exact gcd ℓ_z, cancel it and use the graph of the rational deck involution of the residual quadratic pencil. The graph has $q + 1$ rational points; at most two are fixed, and at most two involve each of z, x_1, x_2, x_3, s. Since $q + 1 > 12$, a graph point remains. In the inseparable quadratic case the characteristic is two and, after a root-coordinate change, the residual pencil is $\langle T^2, U^2 \rangle$. On the affine chart $\ell_z = t - z$, its two kernel cubics have coefficient vectors

$$(-z, 1, 0, 0), \qquad (0, 0, -z, 1).$$

The two overlapping Hankel rows give successively

$$c_1 = zc_0, \quad c_2 = z^2 c_0, \quad c_3 = z^3 c_0, \quad c_4 = z^4 c_0.$$

Hence the determinant (3) vanishes; the homogeneous $z = \infty$ chart gives the endpoint. Thus the inseparable case really lies on the rank-drop carrier. Multiplication by ℓ_s gives a split squarefree quartic avoiding x_1, x_2, x_3.

Two-pointed quintic series. Let $f \in S_6$ carry two distinct markers x_1, x_2 and lie outside the lower persistent, binary-nucleus, fixed-marker, and collision boundary of Definition 7.12; contract at r. The lower rank-two and binary-nucleus pullbacks then cost at most $3 + 1$. If the contracted quartic net has exact gcd ℓ_z, the condition $z = r$ lies in the ramification divisor of a generic separable projection of the moving g^3_d, $d \leq 5$, and costs at most 8. The conditions $z = x_i$ are the two fixed-evaluation incidences (4), each of degree at most two; $r = x_i$ contributes two diagonals. Hence

$$3 + 1 + 8 + 2 \cdot 2 + 2 = 18. \tag{13}$$

The same rank, evaluation-hyperplane, and separable-projection arguments show that no listed pullback is the whole r-line off the declared boundary.

For a trivial-gcd contracted net, the preceding three-pointed lemma applies to x_1, x_2, r. For exact gcd ℓ_z away from these markers, cancel ℓ_z. In the ordered-pair parametrization of the residual cubic hyperplane, avoiding z, x_1, x_2, r gives four bidegree-$(1,1)$ equations; the two root self-collisions have bidegrees $(2,1)$ and $(1,2)$. Their rational zero sets contain at most $14(q+1)$ ordered pairs, whereas

$$(q-3)(q-4) > 14(q+1) \qquad (q \geq 53).$$

Thus a split squarefree quartic remains. The cases $z \in \{x_1, x_2, r\}$ are exactly the excluded fixed-evaluation and ramification divisors. Lifting by ℓ_r gives a split squarefree quintic avoiding x_1, x_2.

One-pointed sextic series. Now let $f \in S_7$ have marker x and lie in the open pointed domain of Definition 7.12; contract at r. The pullback of the lower rank-two carrier has degree at most three. The complete consecutive-row Lucas calculation gives at most two linear nucleus-support pullbacks. A whole binary-central line belongs to the binary Lucas-nucleus component, and the other whole-line possibilities are exactly the remaining declared modular supports, so these pullbacks are finite on the pointed domain. After removing fixed factors, a generic projection of the moving g^4_d, $d \leq 6$, is separable: an inseparable series of degree at most six has dimension at most $\lfloor 6/p \rfloor + 1 < 5$. Its ramification degree is at most $2d - 2 \leq 10$, and it contains the collision in which the new gcd root equals r. The fixed-root incidence at x costs at most two, and $r = x$ costs one. The total is

$$3 + 2 + 10 + 2 + 1 = 18. \tag{14}$$

Choose r off this divisor and apply the two-pointed quintic result with markers x, r. Its quintic avoids both, so multiplication by ℓ_r gives a split squarefree sextic in W_f avoiding x. This is $\mathrm{LP}(7,1)$.

Finally, let $f \in S_8$ lie outside the persistent, modular, and self-collision components of $\mathrm{CC}(8,1)$. Its first-polar line meets the lower rank-two carrier in degree at most three and the complete linear nucleus support in degree at most two. The new-marker collision is ramification of a separable g^5_d, $d \leq 7$: an inseparable series would have dimension at most $\lfloor 7/p \rfloor + 1 < 6$, and Riemann–Hurwitz gives at most $2d - 2 \leq 12$. This proves (11) and excludes an additional contained collision component.

The gcd at every recursive level has degree zero, one, or at least two. The last case lies in the degree-appropriate rank-two catalecticant carrier, whose bottom equation is (3) and whose contained-line pullback is Proposition 5.8; the degree-one case is treated after cancellation. A separable gcd-zero cubic map has geometric monodromy S_3 or C_3, with the latter and every inseparable case in (5)–(6) or the derivative-minor boundary. Hence the strata are exhaustive. Since

$$q + 1 - 2\sqrt{q} > 36$$

at $q = 53$ and the left side is increasing for $q > 1$, (10) leaves a rational point for every $q \geq 53$. The parameter budgets (12)–(14) and (11) are all smaller than $q + 1$. The first prime power satisfying the genus-one inequality is 53. $\square$

The only substantial modular carrier occurs in characteristic seven and is a projective binary-quartic space. Fixing a split quintic $P(t) = \sum_{i=0}^{5} p_i t^i$ and a carrier point $h = (a_0, \ldots, a_4)$, put

$$H_j = \sum_{i=0}^{4} a_i p_{i+j}$$

with $p_k = 0$ outside $0 \leq k \leq 5$, and

$$D = H_0 H_2 - H_1^2, \quad N_s = H_{-1} H_2 - H_0 H_1, \quad N_u = H_{-1} H_1 - H_0^2.$$

Off $D = 0$, the unique residual quadratic is

$$Dt^2 - N_s t + N_u,$$

with branch polynomial $K = N_s^2 - 4N_u D$. The exact residual identities are proved next; the geometric exhaustion of transported slices is kept as a separate printed gate.

Proposition 7.14 (Residual quadratic identities)**.** With $p_k = 0$ outside $0 \le k \le 5$, the two consecutive Hankel equations for

$$P(t)(t^2 - st + u)$$

are

$$H_0 - sH_1 + uH_2 = 0, \qquad H_{-1} - sH_0 + uH_1 = 0.$$

On $D \neq 0$, Cramer's rule gives

$$s = N_s/D, \qquad u = N_u/D.$$

The branch divisor is $K = 0$, fixed/residual collision is $\operatorname{Res}(P, Dt^2 - N_s t + N_u) = 0$, and fixed-root collision is $\operatorname{Disc}(P) = 0$.

Proof. Expand the two consecutive coefficients annihilated by the carrier quartic. The displayed linear system has determinant D, so Cramer's rule gives the residual factor. Its discriminant is K. The two resultants have their usual intrinsic collision meanings; consequently the construction commutes with scalar, PGL_2, and Frobenius transport. $\square$

Proposition 7.15 (One-moving-root slice)**.** Let $P = R(t)(t - x)$ with four fixed distinct roots. Then D, N_s, N_u have degree at most two in x, and K has degree at most four. If the squarefree reduction of K is nonconstant and not a square, the normalization of $y^2 = K(x)$ is geometrically integral of genus at most one.

Proof. Each H_j is affine in x, so its 2×2 minors are quadratic and K is quartic. In odd characteristic the squarefree nonsquare equation defines a geometrically integral double cover; the hyperelliptic genus formula gives $\lfloor (\deg K_{\mathrm{red}} - 1)/2 \rfloor \le 1$. $\square$

Proposition 7.16 (Binary-quartic slices and rational base selection)**.** Every nonzero geometric binary quartic has a four-root slice whose reduced residual cover is geometrically integral of genus at most one. If the quartic is rational over a characteristic-seven field $\mathbb{F}_q$ with $q > 102$, the four roots may be chosen distinct and rational.

Proof. Let

$$\mathcal{H} = \mathbb{P}(\operatorname{Sym}^4 E), \qquad \mathcal{B} = \operatorname{Conf}_4(\mathbb{P}^1).$$

Over $\mathcal{H} \times \mathcal{B}$, the residual formulas define a universal branch quartic $K_{h,\mathbf{r}}(x)$. Its subdiscriminants cut out the locus where the squarefree part drops degree. Hence the good-slice incidence

$$\mathcal{U} = \{(h, \mathbf{r}) : K_{h,\mathbf{r},\mathrm{red}} \text{ is nonconstant and separable}\}$$

is a PGL_2-stable open set. The geometric component problem is the surjectivity of $\mathcal{U} \to \mathcal{H}$. One successful specialization would show only generic nonemptiness; fiberwise surjectivity requires an open cover of quartic moduli.

$$\begin{array}{ccccc} h_L & \xrightarrow{\text{six sections}} & (\Delta_1,\dots,\Delta_6) & \xrightarrow{\sum b_i\Delta_i=1} & \bigcup_i D(\Delta_i)=\mathbb{A}^1_L \\ \{4,\ 3+1,\ 2+2,\ 2+1+1\} & \xrightarrow{\text{disc}=3,3,1,5} & \text{good slices on every boundary stratum.} & & \end{array}$$

Figure 3: The redundancy-nine good-slice argument. The Bézout identity covers the entire squarefree normal-form atlas; four root-partition calculations cover its boundary. The conclusion is fiberwise surjectivity, not generic success of one specialization.

The quotient has one one-dimensional squarefree stratum and four multiple-root boundary strata. On the standard even normal-form atlas for the squarefree stratum, write $h_L = [1, 0, L, 0, 1]$. The six four-point sets and the coefficient vectors of $\Delta_i(L)$ and $b_i(L)$ are printed in Appendix A. They are six test sections of the universal base family over this atlas. Pullback of the branch discriminant gives $\Delta_i(L)$, and direct multiplication gives

$$\sum_{i=1}^{6} b_i(L)\Delta_i(L) = 1 \quad \text{in } \mathbb{F}_7[L]. \tag{15}$$

Thus the principal opens $D(\Delta_i)$ cover the normal-form atlas; these are six sections, not six isolated successful tests. Transport moves the quartic and its four-point base together, so the existence statement descends despite the many-to-one normal form. On the boundary, root multiplicity gives exactly the partitions $4, 3+1, 2+2, 2+1+1$: there are at most three support points, and triple transitivity gives one orbit per partition. Their reduced branch discriminants are $3, 3, 1, 5$, respectively. Thus every boundary orbit also meets $\mathcal{U}$, proving surjectivity.

Now fix $h \in \mathcal{H}(\mathbb{F}_q)$. Over $\mathbb{F}_q(\mathbf{r})$ let $d(h) \geq 1$ be the number of distinct roots of the generic $K_{h,\mathbf{r}}$. Define $B_h(\mathbf{r})$ to be its $d(h)$-th subdiscriminant, equivalently the corresponding nonzero Hermite principal minor. This is a polynomial, not a discriminant after an unspecified gcd division. Its nonvanishing after specialization guarantees a nonconstant separable squarefree part. Surjectivity over the algebraic closure proves $B_h \neq 0$. The affine chart of four finite roots is dense in $\mathcal{B}$, so this nonempty good open meets the chart used below. The bad affine base locus is contained in

$$B_h(\mathbf{r}) \prod_{i<j} (r_i - r_j) = 0, \tag{16}$$

so rational selection is avoidance of a proper divisor in configuration space, not a second monodromy assertion.

Every coefficient of $R_{\mathbf{r}}(t)(t-x)$ is affine in x, of degree at most one in each r_i and total $\mathbf{r}$-degree at most four. Hence D, N_s, N_u have individual degree at most two and total degree at most eight; K has individual degree at most four and total degree at most sixteen. Every quartic subdiscriminant is a Hermite minor of coefficient-degree at most six. Therefore

$$\deg_{r_i} B_h \leq 24, \qquad \deg B_h \leq 96. \tag{17}$$

Multiplying by the four-root Vandermonde gives a nonzero polynomial of total degree at most 102. Schwartz–Zippel leaves a rational ordered base with distinct roots whenever $q > 102$. □

Proposition 7.17 (Good-slice deletion)**.** Once a rational base with an integral slice is fixed, the moving/fixed diagonal, determinant, branch, four fixed-root collisions, and moving-root collision delete at most, respectively,

$$8,\ 4,\ 4,\ 8,\ 8$$

points on the normalized double cover. Their total is 32, so $q+1-2\sqrt{q} > 32$ supplies a split witness.

Proof. The projection of the normalized residual cover to the moving-root line has degree two. Therefore the degree-four fixed-root diagonal and the degree-two determinant pull back to at most eight and four points. The degree-four branch divisor is ramified and contributes at most four points. For each of the four fixed roots the residual-root collision is linear in the moving parameter; its pullback contributes at most two points, for a total of eight. The moving-root resultant has degree at most four and contributes at most eight points after pullback. Removing their union from the normalized genus-at-most-one slice proves the claim. □

Proposition 7.18 (Characteristic-seven finite bridges)**.** At $q = 7$, the formal carrier has exactly 819 projective rootless quartics. At $q = 49$, Certificate R9-49 exhausts all

$$\frac{49^5-1}{48} = 5{,}884{,}901$$

projective carrier points and finds a split witness for each.

Proof. At $q = 7$, a split squarefree degree-seven projective divisor is the complement of one of the eight rational points. Inclusion–exclusion gives

$$\frac{7^5-1}{6} - 8\frac{7^4-1}{6} + 28\frac{7^3-1}{6} - 56\frac{7^2-1}{6} + 70\frac{7-1}{6} = 819.$$

The $q = 49$ clause is deliberately certificate-only: one lexicographic five-root search stops after every carrier point has a witness. The residual algebra is independently replayed, but there is no second exhaustive carrier implementation. □

Proposition 7.19 (R9 persistent orbit law)**.** Put $d = \gcd(8, q+1)$. The sigma-fiber projective orbits are the inversion orbits in $C_d = T/T^8$, and coefficient Frobenius acts by multiplication by the characteristic. The tangent fiber is transitive unless the characteristic is two, when its zero and nonzero parts are separate.

Proof. The nonsplit torus acts by eighth powers and its normalizer inverts the torus coordinate. Thus inversion-fixed and paired classes have stabilizers $2d$ and d, respectively; coefficient Frobenius is the p-power map. For $d = 8$, it fuses the two odd pairs exactly when $p \equiv \pm 3 \pmod 8$. On the tangent fiber the unipotent action is $z \mapsto z + 8u$, which vanishes exactly in characteristic two. □

Proposition 7.20 (Other modular lifts)**.** The consecutive Lucas supports leave a line in characteristic five and the quartic carrier above in characteristic seven; no other characteristic has a nonzero coherent lift. The former line is shallow for every $q > 5$.

Proof. Lucas' theorem and the consecutive-row overlap give the two supports. In characteristic five, homogenize $t^5 - t$ to include its simple root at infinity and multiply by a linear factor $t-a$ with $a \notin \mathbb{F}_5$. The seven roots are distinct and the two required Hankel coefficients vanish. Such an a exists for every $q > 5$. □

Proposition 7.21 (Degree-eight contained components)**.** The assertion $\mathrm{CC}(8,1)$ holds. The ordinary contained component is the upper rank-two carrier. The only modular lifts are the characteristic-five point and the characteristic-seven binary-quartic carrier of Proposition 7.16; the former is shallow for every field in the theorem range. The marked self-collision scheme is never the whole polar line.

Proof. Proposition 5.8 gives the ordinary component, and Proposition 7.20 gives the complete Lucas overlap and the characteristic-five witness.

For collision, remove the fixed divisor from the six-dimensional Hankel series and let its moving degree be $d \leq 7$. If the associated map were inseparable in characteristic p, its sections would lie in the pullback of a series of degree $\lfloor d/p \rfloor$, of dimension at most

$$\lfloor d/p \rfloor + 1 \leq 4 < 6.$$

Thus the moving series is separable. Its ramification divisor has degree at most $(5+1)(7-5) = 12$, so collision is finite and supplies no additional contained component. ☐

Theorem 7.22 (Redundancy nine)**.** For every prime power $q \geq 53$, the deep syndromes of $\mathrm{PRS}(q-8)$ are exactly the persistent tangent and conjugate-secant families, of total size $q(q+1)^2/2$. Their orbit law is T/T^8 modulo inversion and Frobenius; the tangent family splits exactly in characteristic two.

Proof. Proposition 7.13 handles every transverse nonmodular point. Proposition 7.21 leaves the persistent and modular supports. Proposition 7.20 eliminates characteristic five. For characteristic seven, the only powers below 343 are 7 and 49, both below the headline range; at every $q = 7^m \geq 343$, Propositions 7.16 and 7.17 produce a witness. The covering-radius theorem completes the promotion to deep holes. ☐

Proposition 7.18 is a carrier boundary, not part of the headline classification: the $q = 7$ count is not a deep-hole statement for the inadmissible parameter $\mathrm{PRS}(-1)$, and the $q = 49$ pass is not an ambient $\mathbb{P}^8$ census.

8 Characteristic two: the ordered Hessian

In characteristic two the divided binary-cubic discriminant becomes the doubled quadric $(AD+BC)^2$, so its pullback cannot distinguish generic splitting. The correct replacement retains an ordered residual root.

Let $\ell = \mathbb{P}\langle U, V\rangle \subset \mathbb{P}(\Gamma^3 E)$ and write

$$\mathrm{Hess}^{\mathrm{div}}(uU + vV) = N_u(u,v)X^2 + N_s(u,v)XY + D(u,v)Y^2.$$

For

$$U = (A, B, C, D_3), \qquad V = (a, b, c, d),$$

the three coefficient quadratics are

$$\begin{aligned} N_u &= (AC + B^2)u^2 + (Ac + aC)uv + (ac + b^2)v^2, \\ N_s &= (AD_3 + BC)u^2 + (Ad + aD_3 + Bc + bC)uv + (ad + bc)v^2, \\ D &= (BD_3 + C^2)u^2 + (Bd + bD_3)uv + (bd + c^2)v^2. \end{aligned}$$

The ordered-Hessian incidence

$$N_u(u,v)X^2 + N_s(u,v)XY + D(u,v)Y^2 = 0 \tag{18}$$

has bidegree $(2,2)$ and arithmetic genus one. Off $N_s = 0$, its Artin–Schreier class is DN_u/N_s^2.

The parameter space of cubic lines is $G(1, \Gamma^3 E)$. Let

$$\Sigma = \{\mathbb{P}(QE) : [Q] \in \mathbb{P}(\mathrm{Sym}^2 E)\} \subset G(1, \Gamma^3 E)$$

be the Veronese surface of common-quadratic lines. In the ambient cubic space, let $\mathcal{Q}_0 = \{AD + BC = 0\} \subset \mathbb{P}(\Gamma^3 E)$. Its two families of lines give two Fano ruling conics in the

Grassmannian; one is the tangent-conic boundary of Σ, and we call the other $\mathcal{N}$. A root-compatible pullback is the family of lines obtained by fixing a split factor R and varying the final two linear factors in the consecutive Hankel contraction of a syndrome.

$$\begin{array}{cccc} G(1,\Gamma^3 E) \supset & \text{degenerate ordered-Hessian locus} & = & \Sigma \cup \mathrm{Fano}_1(\mathcal{Q}_0) \\ & \downarrow \text{ root-compatible pullback} & & \\ & \text{persistent component} & & \\ & \cup & & \mathcal{N}:\ \text{no rank-two pullback.} \\ & \text{Lucas-nucleus component} & & \end{array}$$

Figure 4: Ordered-Hessian geometry. The degeneracy locus is computed in the Grassmannian first; root compatibility then selects the contained syndrome carriers and eliminates the complementary ruling.

Theorem 8.1 (Ordered-Hessian geometric degeneracy)**.** After removing the forced vertical factors from intersections with the twisted cubic, the complete geometric degeneracy locus of (18) consists of:

(i) the Veronese surface Σ of lines $\mathbb{P}(QE)$, on which the incidence is separably reducible; and

(ii) the two Fano ruling conics of $\mathcal{Q}_0$, on which the root projection of (18) is inseparable.

The complementary ruling $\mathcal{N}$ has no nontrivial rank-two root-compatible Hankel pullback.

Proof. A vertical factor divides N_u, N_s, D exactly when the divided Hessian vanishes at the corresponding point of ℓ. Its base scheme is the twisted cubic, so the vertical factors are precisely $\ell \cap C_3$. A horizontal factor fixes one Hessian root for every cubic on ℓ. The cubics with that root form the rank-three quadric cone swept by secants through the corresponding point of C_3; every line on the cone is a generator through its vertex. Thus a horizontal factor is already a chord/tangent line in Σ, or its collision boundary.

After removing those factors, bidegree additivity forces any separable factorization to be $(1,1)+(1,1)$. Both factors are graphs of projectivities. Make the first graph the identity by independent changes of coordinate on the parameter and root lines. Over an algebraically closed field of characteristic two, the second projectivity is exactly one of the following: the identity; a semisimple element with two fixed points, conjugate to $g(t) = ct$ with $c \neq 1$; or a nontrivial unipotent element with one fixed point, conjugate to $g(t) = t+1$. This is the Jordan classification of 2×2 matrices modulo scalars, so the comparisons below exhaust all pairs of projectivity graphs. For a semisimple second graph $g(t) = ct$, normalize the two cubics at its fixed parameters to

$$U = (a, 1, 0, 0), \qquad V = (0, 0, \gamma, d).$$

Comparison with

$$(uX + vY)(uX + cvY) = u^2X^2 + (1+c)uvXY + cv^2Y^2$$

gives

$$d = 0, \qquad a = 0, \qquad \gamma = 1 + c, \qquad \gamma^2 = c.$$

Hence $c^2 + c + 1 = 0$, and the normalized line is $\mathbb{P}\langle X^2Y, XY^2\rangle = XY \cdot \mathbb{P}(E)$. Undoing the normalization gives exactly $\mathbb{P}(QE)$. In the unipotent normal form $g(t) = t+1$, comparison with

$$(uX + vY)((u+v)X + vY)$$

instead gives $d = \gamma = 0$ and $a\gamma = 1$, a contradiction. The identity case is included in the semisimple comparison and fails the same equations. Thus Σ is the complete separable external locus. The parameter and root coordinate changes used above are precisely basis changes on ℓ and E; the ordered-Hessian incidence and the family $\{\mathbb{P}(QE)\}$ are equivariant for these changes. Undoing the normalization therefore returns a line in Σ and introduces no additional component.

For the inseparable locus, N_s vanishes identically exactly when ℓ is a line on the smooth quadric $\mathcal{Q}_0 = \{AD + BC = 0\}$. Its Fano scheme has the two stated ruling conics. This classifies inseparable root projection, whether or not the total $(2, 2)$ curve is reduced. The exact Plücker equations and the constrained calculation excluding the complementary ruling are given in Proposition 8.2. $\square$

Proposition 8.2 (Root-compatible pullback)**.** On the root-compatible Hankel parameter space, the reduced pullback of Σ is exactly the persistent catalecticant/Lucas-nucleus union. The complementary ruling $\mathcal{N}$ has no nontrivial rank-two pullback. Intersections with the twisted cubic contribute only the vertical factors removed before the reduced incidence is formed.

Proof. Use Plücker coordinates

$$(z_0, \ldots, z_5) = (p_{01}, p_{02}, p_{03}, p_{12}, p_{13}, p_{23}).$$

For $Q = aX^2 + bXY + cY^2$, the line QE has coordinates

$$[z_0 : \cdots : z_5] = [a^2 : ab : ac : b^2 + ac : bc : c^2].$$

Consequently Σ is cut out scheme-theoretically by the 2×2 minors of

$$\begin{pmatrix} z_0 & z_1 & z_2 \\ z_1 & z_3 + z_2 & z_4 \\ z_2 & z_4 & z_5 \end{pmatrix}. \tag{19}$$

The tangent ruling conic is

$$z_1 = z_4 = 0, \quad z_2 = z_3, \quad z_0 z_5 = z_2^2,$$

the square-quadratic boundary of the displayed parametrization. The complementary ruling is

$$\mathcal{N}: \quad z_0 = z_5 = 0, \quad z_2 = z_3, \quad z_1 z_4 = z_2^2. \tag{20}$$

Together with the Chow equation for lines meeting C_3, (19)–(20) are the complete universal degeneracy ideal found in the theorem.

For a split base R, the root-compatible line $\ell_{f,R} = L_f(\mathbb{P}(RE^\vee))$ has Plücker coordinates quadratic in the coefficients of R. If its generic point lies in Σ, the minors (19) say that every iterated contraction pencil has a common quadratic. Proposition 5.8, applied to the overlapping contraction flags, gives the persistent rank-two carrier; when a consecutive contraction coefficient vanishes in the ground characteristic, the same overlap equations and Lucas' theorem give exactly the contraction-kernel nucleus flags. Conversely, the catalecticant and Lucas kernels make every minor in (19) vanish. This proves equality of the reduced pullback with the stated union, not only set-theoretic containment in one direction.

For $\mathcal{N}$, the linear equations in (20), after substitution, give the consecutive identities

$$h_{-1}(R)^2 = h_{-2}(R)h_0(R), \qquad h_1(R)^2 = h_0(R)h_2(R).$$

If these hold for every split R, they hold polynomially because split forms are Zariski dense over the algebraic closure. The contraction forms h_i are linear forms in the coefficients of R. In their UFD, an identity $L_1^2 = L_0L_2$ implies either a zero form or proportionality of the three nonzero forms: every irreducible factor of L_0 or L_2 must be associated to L_1. Applying this to the two overlapping triples propagates proportionality across $h_{-2}, \dots, h_2$. Zero-form cases only lower the same contraction rank. Hence rank $\ell_{f,R} \le 1$, so $\mathcal{N}$ has no nontrivial rank-two pullback.

Finally, the divided Hessian vanishes exactly on C_3. A line–twisted-cubic intersection is therefore exactly the common vertical factor removed above. Scheme-theoretic division by that factor commutes with field extension and cannot create a reduced moving component. □

Proposition 8.3 (Effective Hessian corollary)**.** For syndrome degree $n \ge 5$, outside the persistent/Lucas union an integral reduced slice exists whenever

$$q \ge \min\left\{\frac{(n-4)(n+11)}{2} + 1,\ 9(n-4)\right\}.$$

Its deletion degree is at most $3n-4$. If also $q + 1 - 2\sqrt{q} > 3n - 4$, every split-free syndrome lies in that union.

Proof. For a split base R of degree $n-4$, the root-compatible Plücker coordinates are quadratic in its coefficients. Outside the contained union, choose one nonzero pulled-back equation A of Σ and one nonzero pulled-back equation B of $\mathcal{N}$. Each has individual root degree at most four, and the single product AB vanishes on the whole reduced bad union and has individual root degree at most eight. Vertical factors have already been removed by Proposition 8.2.

Put $m = n-4$. Schwartz–Zippel applied to AB after adjoining the Vandermonde requires

$$q > 8m + \binom{m}{2} = \frac{m(m+15)}{2},$$

which gives the first bound. Alternatively partition $9m$ field elements into disjoint nine-element blocks. Successive univariate interpolation shows that a nonzero polynomial of individual degree at most eight cannot vanish on their product, giving the second bound. The moving diagonal, determinant, branch, fixed-root collision, and final collision divisors have degrees

$$n-4,\ 2,\ 2,\ 2(n-4),\ 4,$$

whose sum is $3n-4$. The normalized integral $(2,2)$ slice has genus at most one, so the point bound leaves a rational ordered residual root outside every deletion, and contraction lifts it squarefreely. □

Theorem 8.1 is a containment theorem. It does not assert that every Lucas-carrier point is deep; that remains an arithmetic question for each level.

9 Lucas carriers and their arithmetic

Let $d = 2^s$ with $s \ge 2$ in characteristic two. Lucas' theorem gives the top nucleus of the degree-d normal rational curve as

$$\mathbb{P}\langle e_1, \dots, e_{d-1}\rangle,$$

whose coherent degree-$(d+1)$ lift is

$$\mathcal{M}_{d+1} = \mathbb{P}\langle e_2, \dots, e_{d-1}\rangle.$$

At the distinguished Borel endpoint e_{d-1} the Hankel kernel contains

$$\mathcal{U}_s = \langle 1, t, t^d \rangle.$$

We use the standard linearized/subspace-polynomial terminology of [LN96, BSEGR16]; the carrier and root-cover calculations below are specific to the present Hankel system.

Proposition 9.1 (Lucas overlap kernel)**.** For $d = 2^s$, the lower nucleus and its coherent lift are exactly

$$\mathbb{P}\langle e_1, \ldots, e_{d-1} \rangle, \qquad \mathbb{P}\langle e_2, \ldots, e_{d-1} \rangle.$$

At e_{d-1}, $\mathcal{U}_s$ is endpoint-specific; the intersection common to the full carrier is only $\langle 1, t^d \rangle$.

Proof. Lucas' theorem gives $\binom{d}{i} \equiv 0 \pmod 2$ for $0 < i < d$. Applying this support condition to the two consecutive contraction rows removes the two endpoint coordinates and gives the displayed lift. For a basis point e_i, the Hankel equations kill the adjacent coefficients b_{i-1}, b_i. At $i = d-1$, the monomials $1, t, t^d$ survive. Varying i over the carrier leaves only $1, t^d$ in the intersection. $\square$

Proposition 9.2 (Linearized root cover)**.** On the generic locus $BC \neq 0$, the normalized ordered-root cover of $A + Bt + Ct^d$ has geometric monodromy $\mathrm{AGL}_1(\mathbb{F}_d)$, minimal constant field $\mathbb{F}_d$, and based deck group C_{d-1}. Coefficientwise Frobenius acts on the based group by multiplication by two, with exact order s. The net contains a split squarefree member over $\mathbb{F}_{2^m}$ if and only if $s \mid m$.

$$K = \overline{\mathbb{F}}_2(a,b) \quad \subset \quad K_1 = K(\beta), \quad \underbrace{\beta^{d-1} = b}_{\substack{\mathbb{F}_d^\times \\ \text{scalings, } d-1}} \quad \subset \quad K_1(y), \quad \underbrace{y^d + y = a/\beta^d}_{\substack{(\mathbb{F}_d,+) \\ \text{translations, } d}}$$

$$(\mathbb{F}_d, +) \rtimes \mathbb{F}_d^\times = \mathrm{AGL}_1(\mathbb{F}_d)$$

Figure 5: The linearized ordered-root cover. The Kummer scaling layer and connected additive layer have coprime roles; their explicit semidirect action accounts for the full geometric monodromy.

Proof. Work first over the geometric coefficient field

$$k = \overline{\mathbb{F}}_2, \qquad K = k(A/C, B/C),$$

on the open set $BC \neq 0$. Put $a = A/C$ and $b = B/C$. The Kummer extension

$$K_1 = K(\beta), \qquad \beta^{d-1} = b,$$

has degree $d-1$: the valuation of b at its simple zero is one. Since k contains $\mathbb{F}_d$, this is the scaling part of the geometric root cover, not an extension of constants. Over K_1 choose

$$y^d + y = a/\beta^d.$$

The roots are $\{\beta(y+c) : c \in \mathbb{F}_d\}$. The right side has a simple pole at the pole of a. If the $\mathbb{F}_d$-torsor were disconnected, its geometric translation group would lie in a proper additive subgroup of $\mathbb{F}_d$. A nonzero $\mathbb{F}_2$-linear functional annihilating that subgroup would then give an Artin–Schreier equation $w^2 + w = \gamma a/\beta^d$ with $\gamma \in \mathbb{F}_d^\times$. This is impossible: the right side has a simple pole, whereas every pole of $w^2 + w$ has even order. Thus $K_1(y)/K_1$ is connected of degree d. Its translations $y \mapsto y + c$ give the normal subgroup $(\mathbb{F}_d, +)$. The $d-1$ Kummer conjugates act by

$$(\beta, y) \longmapsto (\zeta\beta, \zeta^{-1}y), \qquad \zeta \in \mathbb{F}_d^\times,$$

and conjugate translations by multiplication by ζ. We have thus exhibited $d(d-1)$ automorphisms of an extension of degree $d(d-1)$. Consequently the full geometric group, rather than a proper subgroup, is

$$(\mathbb{F}_d, +) \rtimes \mathbb{F}_d^\times = \mathrm{AGL}_1(\mathbb{F}_d).$$

Return now to the arithmetic coefficient field $\mathbb{F}_2(A/C, B/C)$. Ratios of nonzero root differences recover every element of $\mathbb{F}_d$, so any splitting field contains $\mathbb{F}_d$ in its constant field. The preceding regular construction over $\mathbb{F}_d$ introduces no larger constants, proving minimality. Coefficient Frobenius acts on $\mathbb{F}_d^\times$ by $c \mapsto c^2$, and hence has exact order s on the based component group C_{d-1}. Boundary members with $C = 0$ have degree at most one, while those with $B = 0$ have zero derivative; neither is split squarefree of degree d. Thus a split member forces $\mathbb{F}_d \subset \mathbb{F}_{2^m}$, equivalently $s \mid m$. Conversely $t^d + t$ splits simply under that inclusion. The equality $\mathrm{ord}_{2^s-1}(2) = s$ follows from

$$\gcd(2^r - 1, 2^s - 1) = 2^{\gcd(r,s)} - 1.$$

□

At $s = 2$ this recovers the odd/even extension law at redundancies six and seven. The first fresh carrier is $s = 3$, in syndrome degree nine. The $\mathbb{F}_8$-linear section has order-three component Frobenius, but its four quotient directions change the full answer.

Proposition 9.3 (The e_7 orbit)**.** The stabilizer of $e_7 \in \mathbb{P}(\Gamma^9 E)$ is the upper Borel, so its PGL_2-orbit is $\mathbb{P}^1$. At the distinguished point,

$$W_{e_7} = \langle 1, t, t^2, t^3, t^4, t^5, t^8 \rangle.$$

Proof. Lucas parity and extraction of the x^2y^7 coefficient give

$$ge_7 = (ad + bc)^2 \big(b^5 e_2 + b^4 d\, e_3 + bd^4 e_6 + d^5 e_7\big).$$

This is proportional to e_7 exactly when $b = 0$. The apolar action is contragredient, so it transports both W_f and split root divisors. The displayed kernel follows from the two Hankel equations. □

Proposition 9.4 (Framed Artin–Schreier quotient)**.** Order roots $0, 1, a, b, c, d, u, v$ and put

$$S = a + b + c + d, \quad E = ab + ac + ad + bc + bd + cd, \quad h = 1 + S, \quad p = 1 + E + S + S^2.$$

On the open set where the six fixed roots are distinct, $h, p, 1 + h + p$ are nonzero, and $x^2 + hx + p \neq 0$ for $x = a, b, c, d$, the remaining pair is the geometrically integral Artin–Schreier cover

$$y^2 + y = p/h^2.$$

Its rational lifting law is $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$.

Proof. The first elementary-symmetric equation gives $u + v = 1 + S = h$. In the second equation the root 1 contributes $S + u + v = 1$, so

$$0 = E + S(u + v) + uv + S + u + v = E + Sh + uv + 1,$$

and hence $uv = p$. Thus u, v are the roots of $z^2 + hz + p$. The divisor $h = 0$ is exactly their collision divisor. On $h \neq 0$, putting $y = u/h$ gives the displayed Artin–Schreier equation.

It remains to prove that its class is geometric, rather than a constant-field artifact. Write $B = b + c + d$, so $h = a + 1 + B$. Along the prime divisor $h = 0$, the leading coefficient of the order-two pole of p/h^2 is

$$D = 1 + bc + bd + cd + B + B^2,$$

in the residue field $\overline{\mathbb{F}}_2(b, c, d)$. Its derivative $\partial D/\partial b = 1 + c + d$ is a nonzero rational function, so D is not a square there. If $p/h^2 = g^2 + g$, the order-two pole would force g to have a simple pole whose principal coefficient squares to D, a contradiction. The class therefore stays nonzero after algebraic extension of the constants, proving geometric integrality and geometric monodromy C_2.

The remaining open conditions have exact collision meanings: $p \neq 0$ excludes $u, v = 0$, $1 + h + p \neq 0$ excludes $u, v = 1$, and $x^2 + hx + p \neq 0$ excludes $u, v = x$ for $x \in \{a, b, c, d\}$. Over $\mathbb{F}_{2^m}$ the standard Artin–Schreier criterion gives a rational pair precisely when $\mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(p/h^2) = 0$. □

Proposition 9.5 (Additive frame cover)**.** The family

$$t^8 + A_4t^4 + A_2t^2 + A_1t + A_0, \qquad A_1 \neq 0,$$

has a connected affine-frame cover with geometric monodromy $\mathrm{AGL}_3(\mathbb{F}_2)$.

Proof. For an $\mathbb{F}_2$-space U and $w \notin U$,

$$L_{U+\langle w\rangle} = L_U^2 + L_U(w)L_U.$$

Induction gives exponents $8, 4, 2, 1$ and $A_1 = \prod_{v\neq 0} v \neq 0$. An affine frame is an origin plus three independent differences.

Let $\mathcal{F}$ be the open affine frame space of tuples $(a; v_1, v_2, v_3)$ with independent v_i. It is geometrically integral. Sending a frame to the monic polynomial with root set

$$a + \langle v_1, v_2, v_3\rangle_{\mathbb{F}_2}$$

lands in the displayed coefficient family. Conversely, the eight geometric roots of a generic member form an affine three-space: the difference set is the kernel of its additive part, and any root is an origin. Its ordered affine frames are therefore exactly one free orbit of

$$\mathbb{F}_2^3 \rtimes \mathrm{GL}_3(\mathbb{F}_2).$$

Thus the generic fiber has exactly $8|\mathrm{GL}_3(\mathbb{F}_2)| = 1344$ points, every two of them differ by one affine transformation, and the invariant function field is the coefficient field. The normalization is connected and has exact geometric deck group $\mathrm{AGL}_3(\mathbb{F}_2)$. □

Remark 9.6 (Logical boundary). The generic Artin–Schreier cover describes the full framed quotient, and Proposition 9.5 identifies a proper additive subcover. The shallowness theorem below does not require a rational point on the generic cover: its witnesses are constructed directly from rational three-dimensional subspaces.

Theorem 9.7 (Shallowness of the degree-nine e_7 orbit)**.** For every $m \geq 3$, the entire PGL_2-orbit of e_7 is shallow over $\mathbb{F}_{2^m}$. The number of monic additive-affine witnesses is

$$\frac{q(q-1)(q-2)(q-4)}{1344}.$$

Proof. Every three-dimensional $\mathbb{F}_2$-subspace $V \subset \mathbb{F}_{2^m}$ gives

$$L_V(t) = \prod_{v \in V}(t - v) = t^8 + A_4 t^4 + A_2 t^2 + A_1 t.$$

The t^7, t^6 coefficients vanish, so $L_V \in W_{e_7}$, and $A_1 \neq 0$ makes its roots simple. Every admissible binary field in the statement contains such a three-space. Proposition 9.3 transports the witness around the orbit.

There are

$$\binom{m}{3}_2 = \frac{(q-1)(q-2)(q-4)}{168}$$

three-spaces. Each has $q/8$ affine cosets, and $L_{a+V}(t) = L_V(t) + L_V(a)$. A split polynomial recovers its root coset, whose difference set recovers V; hence distinct cosets or three-spaces cannot be counted twice. Multiplying the two factors gives the displayed formula. □

Thus the order-three law of Proposition 9.2 belongs to a proper $\mathbb{F}_8$-linear section, not to deepness of the full e_7 kernel. The other intrinsic strata of the six-dimensional degree-nine carrier are not classified here.

10 Verification and formal boundary

The conceptual theorems above are accompanied by deterministic evidence bundles. Table 8 records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table 2 gives the corresponding claim-level mathematical dependencies.

Table 8: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen finite fields	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate
Certificate R8	fixed-level characteristic cases and bounds	independent algebra/nucleus replay; not an ambient census
Certificate R9	residual algebra and transported slices	independent residual/slice replay; geometric integrality remains a paper proof
Certificate R9-49	characteristic-seven carrier at $q = 49$	one exhaustive carrier implementation; not a whole-code census
Certificate Hessian	bounded ordered-Hessian algebra	regression only; does not replace the geometric component proof
Certificate Lucas	recorded Lucas parameter domain	independent arithmetic replay
Certificate e7	quotient-cover open set and additive specialization	independent quotient-cover replay

The public labels are defined by `supplement/CERTIFICATE-SCHEMA.md`. The paper-local bundle, literal replay commands, toolchain locks, hashes, byte counts, and working directories are recorded in `supplement/REPRODUCING.md`, `supplement/EVIDENCE-MANIFEST.json`, and `supplement/RELEASE-MANIFEST.md`. The single entry point `python3 supplement/verify.py` checks the bundle manifest, classification records, and their expected hashes; its `--replay` option runs the paper-local Python replays.

Lean formalizes the redundancy-nine residual identities, the witness-to-shallow synthesis implication, and the four-marker integer budgets. It does not formalize geometric integrality, Hasse–Weil existence after deletion, the $\mathrm{PGL}_2/\mathrm{P\Gamma L}_2$ action, identification with PRS syndrome families, or geometric exhaustion; those remain printed mathematical inputs. The exact declarations, import gates, source anchors, and axiom-audit output are listed in `supplement/LEAN-STATEMENTS.md`.

Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. Section 10 assigns each computational or formal claim its exact trust route.

AI systems were used under the author's direction for proof exploration, symbolic and finite computation, code generation, and drafting assistance. The author reviewed the resulting arguments and computations and assumes responsibility for every claim in the manuscript. AI systems are not authors.

11 Scope and open problems

The results proved here do not constitute an arbitrary-redundancy classification.

1. Redundancy five is complete for every $q \geq 7$.
2. Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.
3. Redundancies eight and nine are complete for $q \geq 43$ and $q \geq 53$, respectively. No bounded-field completion is inferred.
4. The characteristic-two ordered-Hessian theorem classifies the geometric degeneracy strata, root-compatible pullback, and effective base-selection bound. It does not classify arithmetic deepness on every Lucas carrier.
5. The e_7 orbit is only one intrinsic stratum of the degree-nine Lucas carrier. The other strata, and therefore a redundancy-ten synthesis, remain open.
6. The effective polar-induction theorem requires a new level-specific monodromy/component calculation at each redundancy. A field census cannot replace this input.

The exact bounded-field census gap is finite but presently uncomputed. For redundancy eight it consists of

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41\};$$

for redundancy nine, after excluding the inadmissible $q = 7, 8$ parameters, it consists of

$$q \in \{9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49\}.$$

The existing R8 and R9 certificates do not enumerate these ambient syndrome spaces. A direct scan at the upper endpoints would visit

$$\#\mathbb{P}^7(\mathbb{F}_{41}) = 199623130728, \qquad \#\mathbb{P}^8(\mathbb{F}_{49}) = 33925283289801$$

directions. Even quotienting by the maximum possible $\mathrm{PGL}_2(q)$ orbit size leaves at least 2898130 and 288480301 orbit representatives, respectively. Thus a bounded completion requires a new canonical-orbit enumerator, measured resource bounds, and an independent replay; it is not obtained by merely extending the current regression field list. The R9-49 computation closes only the characteristic-seven carrier named in Section 10, not the ambient $q = 49$ row.

Thus the induction theorem is not an arbitrary-redundancy classification, the redundancy-eight and redundancy-nine results are high-field theorems, and bounded-field and Lucas-carrier strata remain open. In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture.

Conjecture 11.1 (Stable polar classification)**.** There is an absolute constant C such that, at redundancy r, every split-free projective syndrome over $\mathbb{F}_q$ with $q \geq Cr^2$ lies either in a persistent rank-two carrier or in a nonshallow component of the level-r Lucas contraction kernel. Equivalently, transverse arithmetic exceptions disappear above a quadratic field threshold. In every characteristic for which that Lucas kernel has no nonshallow component, the persistent tangent and conjugate-secant families are the complete high-field list.

The Lucas clause is essential: the binary nucleus families at redundancies six and seven persist over infinitely many fields, so a "persistent families only" conjecture would already be false. Conjecture 11.1 predicts instead that the two mechanisms isolated in this paper—rank-two persistence and digit-controlled contraction kernels—exhaust the stable locus, with the polar flag eliminating everything transverse.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; the modular geometry is a Lucas-computed contraction kernel; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while the later theorems show how that cover is propagated, contained, or replaced in the levels and carrier strata treated here.

A Explicit redundancy-nine slice cover

This appendix prints the finite polynomial data used in Proposition 7.16. All coefficients lie in $\mathbb{F}_7$. For a vector $[c_0, \ldots, c_m]$, write

$$[c_0, \ldots, c_m](L) = \sum_{j=0}^{m} c_j L^j.$$

On the squarefree quartic atlas $h_L = [1, 0, L, 0, 1]$, the following ordered four-point bases give the residual branch discriminants $\Delta_i(L)$.

i	fixed roots	coefficients of Δ_i, constant term first
1	`[0, 1, 2, 3]`	`[6, 0, 6, 0, 3, 4, 5, 5, 0, 0, 2, 6, 1, 3, 3, 0, 0, 4, 5, 2, 6, 0, 0, 1, 1]`

i	fixed roots	coefficients of Δ_i, constant term first
2	[0, 1, 2, 4]	[0, 0, 4, 5, 1, 5, 2, 3, 1, 6, 1, 5, 4, 6, 0, 3, 0, 0, 4, 1, 2, 0, 6, 0, 4]
3	[0, 1, 2, 5]	[2, 6, 2, 1, 1, 3, 0, 1, 6, 5, 0, 2, 3, 4, 4, 4, 5, 4, 0, 4, 6, 3, 1, 5, 5]
4	[0, 1, 2, 6]	[0, 0, 0, 0, 0, 0, 3, 0, 0, 6, 5, 0, 0, 1, 0, 0, 2, 4, 0, 0, 3, 0, 0, 6, 5]
5	[0, 1, 3, 4]	[6, 1, 3, 4, 6, 5, 4, 6, 6, 4, 0, 0, 2, 6, 1, 0, 1, 2, 5, 4, 1, 5, 4, 3, 5]
6	[1, 2, 3, 4]	[6, 5, 5, 6, 0, 4, 0, 5, 6, 3, 6, 6, 0, 3, 4, 1, 4, 2, 3]

The Bézout coefficients in (15) are:

i	coefficients of b_i, constant term first
1	[6, 4, 1, 2, 3, 5, 5, 1, 6, 5, 0, 0, 4, 5, 3, 6, 2, 1, 1, 4, 3, 1, 2, 6, 4, 2, 6, 4, 6, 3, 0, 5, 5, 2, 6]
2	[1, 0, 5, 1, 1, 5, 4, 2, 6, 5, 0, 4, 6, 4, 4, 4, 1, 3, 0, 1, 3, 3, 5, 1, 6, 4, 3, 6, 5, 0, 2, 4, 4, 5, 2]
3	[]
4	[]
5	[]
6	[0, 3, 5, 4, 0, 1]

Only b_1, b_2, b_6 are nonzero, but all six sections are retained as regression controls. Direct multiplication in $\mathbb{F}_7[L]$ gives (15), so the principal opens $D(\Delta_i)$ cover the squarefree atlas.

For completeness, the multiple-root boundary has the following four normal forms. The branch vector is again written constant term first.

partition	h	fixed roots	reduced branch	discriminant
4	$[1, 0, 0, 0, 0]$	$[1, 2, 3, 4]$	$[2, 2, 1]$	3
$3+1$	$[0, 1, 0, 0, 0]$	$[0, 2, 4, 6]$	$[4, 4, 3]$	3
$2+2$	$[0, 0, 1, 0, 0]$	$[0, 1, 2, 3]$	$[5, 5, 2, 1]$	1
$2+1+1$	$[0, 1, 6, 0, 0]$	$[0, 1, 2, 3]$	$[3, 5, 6, 6, 6]$	5

Every reduced discriminant is nonzero. The squarefree atlas and these four boundary strata therefore exhaust all nonzero geometric binary quartics used in the surjectivity argument.

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